



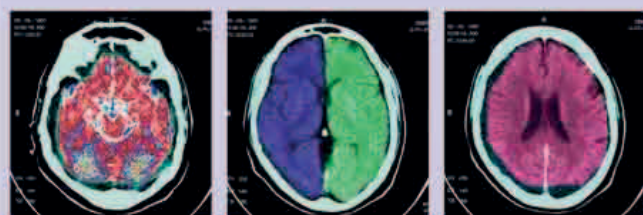
NASA's Apollo Lunar Roving Vehicle (LRV) is driven on the surface of the Moon. During the Apollo 15 mission in July, the first LRV traveled a total of nearly 17 miles (28 km).

THE FIRST ALL-ELECTRONIC CALCULATORS appeared in the 1950s, but they were large and expensive. In the 1960s, calculators became smaller and cheaper, thanks to the use of integrated circuits, or "chips"—tiny but complex electronic circuits etched onto a single piece of semiconductor material. In 1968, adding-machine company **Busicom** was designing a portable calculator, and approached two chip manufacturers: Intel and Mostek. Intel developed a chip with the entire central processing unit of a computer in its tiny circuits. It was the **world's first commercially available "microprocessor"** (see panel, right), the Intel 4004. Busicom chose not to use the Intel 4004, because it would have made the cost of producing the calculator prohibitively high. It chose a



simpler chip, made by Mostek. The **first truly pocket-sized calculator**, the LE-120A **HANDY**, went on sale in January 1971. In May, US computer company IBM announced a new convenient data storage device—the **floppy disk**. The floppy played a key role in enabling the development and popularity of **personal computers** because users could store electronic documents, carry them from computer to computer, and send them

First pocket calculator
Busicom's **HANDY LE**, had a red LED display. It featured a fixed decimal point, with a choice of four, two, or no decimal places. It cost \$395.



COLORED CT SCANS OF THE HUMAN BRAIN

CT SCANNING

In computed tomography (CT), an X-ray source and a detector are placed on opposite sides of a rotating drum. A person lies inside the drum and an X-ray beam passes through him or her. The signal in the detector depends on the average density of the material through which the beam passes. As the drum rotates, a computer combines multiple scans, to produce a 2-D "slice" through the person.

through the mail. Floppy disks were thin plastic disks coated with magnetic particles, and were attached to a spindle inside a rigid plastic housing. The first floppy disks were 8 in (20.3 cm) in diameter, and **held 80 kB (kilobytes) of information**.

Another important first in 1971 was e-mail (electronic mail). For

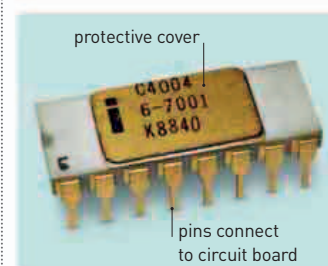
about a decade, different users logged into the same computer were able to leave messages for each other. With the creation of the US military's ARPANET network in 1969, organizations **could send information across large distances**. American computer programmer Ray Tomlinson designed an e-mail system for ARPANET and **sent the first true e-mail**.

In space, the USSR's **Salyut 1** became the **first ever orbiting space station**, remaining in orbit for 175 days. The cosmonauts of the Soyuz 11 mission were the first to enter the craft, in June. They spent three weeks inside, but all three were killed as they were leaving; they remain the only people to have died outside Earth's atmosphere.

In July, astronauts from NASA's Apollo 15 mission drove around the lunar surface in the first **Lunar Roving Vehicle**, or Rover.

In medical science, the first commercially available CT (computed tomography) scanner

was released—and the first scans of the human brain produced. American inventor Raymond Damadian published the results of his experiments into **magnetic resonance imaging (MRI)** (see 1977). British pharmacologist John Vane published his research that **explained how the painkiller aspirin works**—by blocking the production of prostaglandins, compounds that play a central role in the mechanism of pain and the body's inflammatory response.



FIRST MICROPROCESSOR

A microprocessor is a single-chip central processing unit (CPU). Beneath the cover is a single flat crystal of silicon onto which thousands, millions, or, more recently, billions of transistors and other components are etched. Microprocessors are one of the defining components of the microelectronics industry.

“ I THOUGHT ABOUT OTHER SYMBOLS, BUT @ DIDN'T APPEAR IN ANY NAMES, SO IT WORKED. ”

Ray Tomlinson, US computer programmer and inventor of e-mail, 1998

January Busicom releases the first pocket calculator, the LE-120A

February 5 Apollo 14 arrives on the Moon; US astronaut Alan Shepard hits golf balls on the lunar surface

April 19 Salyut 1 becomes the first ever orbiting space station

April 22 MMR (measles, mumps, and rubella) combination vaccine licensed in the USA

June 30 The three-member crew of the Soyuz 11 spacecraft die shortly after exiting the Salyut 1 space station

July 30 Apollo 15, the fourth manned mission to the Moon, arrives on the lunar surface

October 1 First human brain scan is carried out with the first commercially available CT scanner

October 4 "Mole," a chemistry unit of measurement, added as standard unit, part of the SI system

Fall American computer programmer Ray Tomlinson sends first-ever e-mail

November 14 NASA's Mariner 9 becomes the first spacecraft to go into orbit around planet Mars

November 15 Electronic News announces Intel's 4004 chip—the world's first microprocessor